

Coherent Green Functions from Declared Projection Kernels

Elliptic Results and the MTT Source Boundary

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Abstract

The preceding paper identified Dirac delta functions as singular shadows of admissible projection: a delta is the zero-width limit of a bounded coherent projection kernel. The present paper develops the first worked sequel. We replace the standard point-source Green equation

$$LG(x, y) = \delta(x - y)$$

by the coherent Green equation

$$LG_{\text{coh}}(x, y) = K_{\text{coh}}(x, y),$$

where K_{coh} is a bounded admissible identity kernel on a retained coherent sector. The mathematical core is elementary but important: for a positive self-adjoint elliptic operator L , finite spectral projectors and heat-kernel filters produce smooth finite-width sources, smooth responses, finite diagonal values, and distributional convergence to the ordinary Green kernel only in the singular limit. A worked circle model with $L_m = -\partial_\theta^2 + m^2$ displays the replacement explicitly: $L_m G_N = K_N$ and $L_m G_\tau = H_\tau$, with the point-source equation recovered only as $N \rightarrow \infty$ or $\tau \downarrow 0$. Thus the standard Green function is recovered, but as an ideal endpoint rather than a primitive object.

Within Modal Triplet Theory (MTT), these theorems provide a conditional downstream interface rather than a universal replacement rule. A selected physical realization must still supply L , the projector or filter, its width, and the map from an internal or Euclidean spectral problem to the claimed external source. It must also prove that covariance, gauge constraints, locality or causal support, and boundary conditions survive. Different finite kernels can share the same sharp limit while giving different finite predictions. The rigorous result is therefore the elliptic response theorem and its singular limit; physical kernel selection remains open.

Version 2 Revision Note

Supersedes Version 1. The previous release remains public until a new release is approved.

Reason Correct boundary-domain and Sobolev-norm claims for Green responses.

Resolution Repairs both convergence theorems and distinguishes spectral from geometric Sobolev norms.

Retained result Correctly scoped results and explanatory examples remain; no physical source-selection claim is promoted.

Remaining boundary Realization hypotheses and physical source or apparatus bridges remain separate obligations. This is an unreleased authoring revision.

Version 1 Revision Note

Supersedes The unversioned manuscript *Coherent Green Functions: Replacing Point Sources by Admissible Kernels in Modal Triplet Theory*.

Reason The earlier text correctly proved spectral and heat-kernel limits but treated a declared finite kernel as physically selected and extended compact elliptic conclusions to point particles, retarded propagators, contact interactions, and renormalization without the necessary source and symmetry theorems.

Resolution This version retains the spectral mathematics, types the result as an elliptic or Euclidean construction, distinguishes a sector identity from the full identity, and makes covariance, gauge, locality, causal-support, and boundary compatibility explicit gates.

Retained result For a supplied positive self-adjoint elliptic L , finite spectral and heat-filtered Green operators are smooth finite-source responses and converge distributionally to the ordinary Green kernel in the sharp limit.

Remaining boundary No theorem here selects the finite kernel or its width from MTT geometry, and the hyperbolic, gauge, and interacting extensions remain separate constructions.

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1 Purpose and claim discipline

The previous paper established the following structural thesis:

$$\delta(x - y) = \text{singular limit of coherent projection kernels.}$$

The present paper asks for the first concrete physical construction that follows from that thesis.

The natural starting point is the Green equation. In standard continuum physics, the response of a linear operator L to a point source is encoded by

$$LG(x, y) = \delta(x - y). \quad (1.1)$$

This formula is mathematically powerful, but it builds in an idealization: the source has zero spatial width and exact point support.

The MTT replacement is:

$$LG_{\text{coh}}(x, y) = K_{\text{coh}}(x, y), \quad (1.2)$$

where K_{coh} is the kernel of a bounded coherent projection or admissibility filter.

Non-claims

This paper does not claim that ordinary Green functions are wrong. It does not claim that all point-source methods should be abandoned. It does not compute numerical finite-width corrections for a specific experimental system, select a physical kernel from MTT geometry, or derive a Lorentzian retarded replacement from the compact elliptic calculation. Its narrower claim is:

The point-source Green function is the singular limit of coherent finite-source Green functions.

This is proved in standard spectral settings and then interpreted inside MTT.

2 Standard Green functions and the hidden delta

Let X be a compact smooth Riemannian manifold without boundary, or a compact domain with self-adjoint elliptic boundary conditions. Let L be a positive self-adjoint elliptic operator of order two on $L^2(X)$. For concreteness one may take

$$L = \Delta + m^2, \quad m^2 > 0,$$

where $\Delta \geq 0$ is the nonnegative Laplacian.

Let $\{\phi_n\}_{n=0}^\infty$ be an orthonormal eigenbasis:

$$L\phi_n = \mu_n\phi_n, \quad 0 < \mu_0 \leq \mu_1 \leq \cdots, \quad \mu_n \rightarrow \infty.$$

The standard Green kernel is formally

$$G(x, y) = \sum_{n=0}^{\infty} \frac{\phi_n(x) \phi_n^*(y)}{\mu_n}. \quad (2.1)$$

It satisfies

$$L_x G(x, y) = \delta(x - y) \quad (2.2)$$

in the distributional sense.

The key point is that the singularity of G is not only a property of L^{-1} . It is also a property of the source:

$$\delta(x - y).$$

If the source is replaced by a finite-width kernel, the response is correspondingly regularized.

3 Coherent source kernels

Definition 3.1 (Spectral coherent source kernel). For $\Lambda > 0$, define the spectral projector

$$\Pi_{\Lambda} f = \sum_{\mu_n \leq \Lambda} \langle \phi_n, f \rangle \phi_n.$$

Its kernel is

$$K_{\Lambda}(x, y) = \sum_{\mu_n \leq \Lambda} \phi_n(x) \phi_n^*(y). \quad (3.1)$$

For finite Λ , K_{Λ} is a finite-rank smooth kernel. It is not the full identity kernel on $L^2(X)$. It is the identity kernel only on the retained spectral subspace

$$\text{Ran } \Pi_{\Lambda}.$$

Definition 3.2 (Heat coherent source kernel). For $\tau > 0$, define the heat kernel of Δ by

$$H_{\tau}(x, y) = \sum_{n=0}^{\infty} e^{-\tau \lambda_n} \phi_n(x) \phi_n^*(y),$$

where $\Delta \phi_n = \lambda_n \phi_n$. This is a positive smoothing approximate identity.

The sharp spectral kernel K_{Λ} and the heat kernel H_{τ} play different roles:

$$K_{\Lambda} = \text{band-limited identity kernel},$$

while

$$H_{\tau} = \text{positive smoothing identity kernel}.$$

Both converge to $\delta(x - y)$ in a singular limit, but only H_{τ} is automatically positive and smoothing in the probabilistic sense.

4 Coherent Green functions

Definition 4.1 (Spectral coherent Green function). The spectral coherent Green function is

$$G_\Lambda(x, y) := (L^{-1}\Pi_\Lambda)(x, y) = \sum_{\mu_n \leq \Lambda} \frac{\phi_n(x)\phi_n^*(y)}{\mu_n}. \quad (4.1)$$

It satisfies the finite-source equation

$$L_x G_\Lambda(x, y) = K_\Lambda(x, y). \quad (4.2)$$

Definition 4.2 (Heat coherent Green function). For a heat source H_τ , define

$$G_\tau(x, y) := (L^{-1}H_\tau)(x, y). \quad (4.3)$$

If $L = \Delta + m^2$ and $H_\tau = e^{-\tau\Delta}$, then

$$G_\tau = L^{-1}e^{-\tau\Delta}. \quad (4.4)$$

It satisfies

$$L_x G_\tau(x, y) = H_\tau(x, y). \quad (4.5)$$

5 Main theorem: coherent Green functions converge to the point-source Green function

Theorem 5.1 (Spectral coherent Green convergence). *Let L be a positive self-adjoint elliptic operator of order two on a compact smooth domain as above. Let $G_\Lambda = L^{-1}\Pi_\Lambda$. Then:*

1. *For each finite Λ , $G_\Lambda(x, y)$ is smooth on $X \times X$.*
2. *$L_x G_\Lambda(x, y) = K_\Lambda(x, y)$.*
3. *As $\Lambda \rightarrow \infty$, $G_\Lambda \rightarrow G = L^{-1}$ in the distributional kernel sense.*
4. *For every $f \in D_\infty(L) := \bigcap_{m \geq 0} \text{Dom}(L^m)$,*

$$G_\Lambda f := \int_X G_\Lambda(x, y)f(y) dy \longrightarrow Gf := L^{-1}f$$

in $C^\infty(X)$. On a closed manifold this includes every smooth input. With boundary, all powers of the selected elliptic realization impose compatibility; arbitrary smooth boundary data are not included. For every $f \in L^2(X)$, convergence still holds in L^2 .

Proof. Because L is positive, self-adjoint, elliptic, and defined on a compact domain with self-adjoint boundary conditions, it has a complete orthonormal eigenbasis $\{\phi_n\}$ with eigenvalues $\mu_n \rightarrow \infty$. The finite spectral sum defining G_Λ is smooth, proving (1).

Applying L_x termwise gives

$$L_x G_\Lambda(x, y) = \sum_{\mu_n \leq \Lambda} \phi_n(x)\phi_n^*(y) = K_\Lambda(x, y),$$

proving (2).

Let $f = \sum f_n \phi_n$. Then

$$G_\Lambda f = \sum_{\mu_n \leq \Lambda} \frac{f_n}{\mu_n} \phi_n, \quad Gf = \sum_{n=0}^{\infty} \frac{f_n}{\mu_n} \phi_n.$$

For $s \geq 0$, define the spectral graph norm $\|h\|_{\mathcal{H}_L^s} := \|(I + L)^{s/2}h\|_2$. Then exactly

$$\|Gf - G_\Lambda f\|_{\mathcal{H}_L^s}^2 = \sum_{\mu_n > \Lambda} (1 + \mu_n)^s \frac{|f_n|^2}{\mu_n^2}.$$

For $f \in D_\infty(L)$ this tail tends to zero at every order. Geometric Sobolev norms are compared by elliptic estimates; they are not identically the displayed spectral norm. Sobolev embedding gives the smooth limit. For general L^2 input use $s = 0$ and boundedness of L^{-1} . The distributional kernel limit follows by testing against smoothing trace-class operators. This is the Green-response specialization of the joint sharp-limit theorem in [1], not an equivalence between smooth and distributional convergence. $\square$

Corollary 5.2 (Coherent Green functions are regularized point-source responses). *For finite Λ , the equation*

$$LG_\Lambda = K_\Lambda$$

is a smooth finite-source response. The ordinary point-source equation

$$LG = \delta$$

is recovered only in the singular limit $\Lambda \rightarrow \infty$.

6 Heat-kernel version

The spectral cutoff theorem is useful because it directly matches finite coherent subspaces. The heat-kernel version is often physically cleaner because the source is positive and localized.

Theorem 6.1 (Heat coherent Green convergence). *Let $L = \Delta + m^2$ with $m^2 > 0$ on a compact Riemannian manifold or compact domain with self-adjoint boundary conditions. Define*

$$G_\tau = L^{-1}e^{-\tau\Delta}.$$

Then:

1. $G_\tau(x, y)$ is smooth for every $\tau > 0$;
2. $L_x G_\tau(x, y) = H_\tau(x, y)$;
3. as $\tau \downarrow 0$, $G_\tau \rightarrow G = L^{-1}$ distributionally;
4. for every $f \in D_\infty(\Delta)$, $G_\tau f \rightarrow Gf$ in $C^\infty(X)$; for every L^2 input, convergence holds in L^2 . The all-powers domain uses the same boundary realization for Δ and $L = \Delta + m^2$.

Proof. Since $e^{-\tau\Delta}$ is smoothing for $\tau > 0$, $G_\tau = L^{-1}e^{-\tau\Delta}$ has a smooth kernel. The identity $L_x G_\tau = H_\tau$ follows from applying L to $L^{-1}H_\tau$.

In the eigenbasis of Δ , one has

$$G_\tau f = \sum_{n=0}^{\infty} \frac{e^{-\tau\lambda_n}}{\lambda_n + m^2} f_n \phi_n, \quad Gf = \sum_{n=0}^{\infty} \frac{1}{\lambda_n + m^2} f_n \phi_n.$$

For $f \in D_\infty(\Delta)$, dominated convergence in each spectral graph norm followed by elliptic Sobolev comparison proves the smooth limit. The order-zero estimate proves the general L^2 limit and trace-class testing proves the distributional kernel assertion. $\square$

7 Finite diagonal values and coincidence singularities

One of the main reasons point-source Green functions create difficulties in QFT and classical field theory is the diagonal limit $G(x, x)$. For a genuine point-source Green function, this is often singular. For coherent Green functions it is finite.

Proposition 7.1 (Finite diagonal for finite spectral response). *For finite Λ ,*

$$G_\Lambda(x, x) = \sum_{\mu_n \leq \Lambda} \frac{|\phi_n(x)|^2}{\mu_n} \quad (7.1)$$

is finite for every $x \in X$.

Proof. The sum contains only finitely many smooth terms. $\square$

Remark 7.2. As $\Lambda \rightarrow \infty$, this diagonal may diverge depending on the dimension and the operator. The divergence is not mysterious in the present framework: it is the cost of driving a finite coherent response toward a zero-width point source.

This gives a precise version of the renormalization intuition introduced in the preceding paper:

Coincident-point singularities arise when finite coherent kernels are forced to the delta limit.

8 Worked model on the circle

We now give an explicit model in which all objects can be written down. This section is not needed for the general theorem, but it makes the replacement rule concrete.

Let

$$X = S^1 = \mathbb{R}/2\pi\mathbb{Z}$$

with coordinate θ , and let

$$L_m = -\frac{d^2}{d\theta^2} + m^2, \quad m > 0.$$

The normalized Fourier modes are

$$e_n(\theta) = \frac{1}{\sqrt{2\pi}} e^{in\theta}, \quad n \in \mathbb{Z},$$

and

$$L_m e_n = (n^2 + m^2) e_n.$$

8.1 Band-limited source kernel

For $N \in \mathbb{N}$, define the finite spectral projector

$$\Pi_N f = \sum_{|n| \leq N} \langle e_n, f \rangle e_n.$$

Its kernel is the Dirichlet spectral kernel

$$K_N(\theta, \eta) = \frac{1}{2\pi} \sum_{|n| \leq N} e^{in(\theta - \eta)} = \frac{\sin((N + \frac{1}{2})(\theta - \eta))}{2\pi \sin((\theta - \eta)/2)}$$

away from $\theta = \eta$, with the removable diagonal value

$$K_N(\theta, \theta) = \frac{2N+1}{2\pi}.$$

The ordinary identity kernel on S^1 is recovered only distributionally:

$$K_N(\theta, \eta) \rightarrow \delta(\theta - \eta)$$

as $N \rightarrow \infty$.

The associated coherent Green kernel is

$$G_N(\theta, \eta) = L_m^{-1} K_N(\theta, \eta) = \frac{1}{2\pi} \sum_{|n| \leq N} \frac{e^{in(\theta-\eta)}}{n^2 + m^2}.$$

It satisfies the exact finite-source equation

$$L_m G_N(\theta, \eta) = K_N(\theta, \eta).$$

Thus G_N is the response to a band-limited source, not to a point source.

8.2 Limit to the ordinary Green function

The ordinary periodic massive Green kernel is

$$G(\theta, \eta) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \frac{e^{in(\theta-\eta)}}{n^2 + m^2}.$$

Equivalently, writing

$$r = d_{S^1}(\theta, \eta) \in [0, \pi],$$

one has

$$G(\theta, \eta) = \frac{\cosh(m(\pi - r))}{2m \sinh(\pi m)}.$$

The finite kernels converge to the ordinary Green kernel:

$$G_N(\theta, \eta) \rightarrow G(\theta, \eta)$$

uniformly in (θ, η) , and distributionally after applying L_m :

$$L_m G_N = K_N \rightarrow \delta.$$

This gives a precise interpretation:

point-source Green response = limit of band-limited coherent Green responses.

8.3 Heat-filtered positive source

The sharp spectral cutoff K_N is finite-rank and smooth, but it is not positive and has oscillatory side-lobes. A smoother positive approximate identity is obtained from the heat kernel:

$$H_\tau(\theta, \eta) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} e^{-\tau n^2} e^{in(\theta-\eta)}, \quad \tau > 0.$$

The corresponding heat-filtered Green kernel is

$$G_\tau(\theta, \eta) = \frac{1}{2\pi} \sum_{n \in \mathbb{Z}} \frac{e^{-\tau n^2} e^{in(\theta-\eta)}}{n^2 + m^2}.$$

It satisfies

$$L_m G_\tau(\theta, \eta) = H_\tau(\theta, \eta),$$

and

$$H_\tau \rightarrow \delta, \quad G_\tau \rightarrow G$$

as $\tau \downarrow 0$.

This explicitly displays the two versions of the replacement:

$$\delta \rightsquigarrow K_N$$

for a band-limited sector identity, and

$$\delta \rightsquigarrow H_\tau$$

for a positive finite-width source.

8.4 What the example shows

The S^1 model illustrates the central claim in a completely explicit setting:

$$LG = \delta$$

is the singular endpoint of the family

$$LG_N = K_N$$

or

$$LG_\tau = H_\tau.$$

The finite kernels are not approximations in a vague philosophical sense. They are exact Green responses to exact finite-resolution sources. The usual point-source equation is recovered only after taking the singular limit.

In one dimension the massive Green function itself has a finite diagonal value, so this example should not be used as a model of coincident-point ultraviolet divergence. Its role is narrower and cleaner: it shows explicitly how the point source is replaced by a coherent finite source, and how the ordinary delta source is recovered as a limit. In higher dimensions, the same replacement also regularizes diagonal singularities before the limit is taken.

9 MTT interpretation and source boundary

Within MTT, the coherent projector Π_{coh} is not normally an increasing full-space spectral cutoff. It is a sector projector: it selects the admissible coherent sector determined by the fixed-point regime, spectral gap, and admissibility conditions.

Thus K_{coh} should not be confused with a literal approximation to the identity on all of $L^2(X)$. It is an identity kernel only inside the retained coherent sector. Moreover, calling that sector physical requires a selected operator, projector, boundary condition, and intertwiner from the relevant MTT carrier.

Definition 9.1 (MTT coherent Green equation). Given a coherent sector with projection kernel K_{coh} , the MTT coherent Green function is defined by

$$LG_{\text{coh}}(x, y) = K_{\text{coh}}(x, y), \quad (9.1)$$

or equivalently

$$G_{\text{coh}} = L^{-1}\Pi_{\text{coh}} \quad (9.2)$$

whenever L^{-1} is defined on the sector under consideration.

The ordinary point-source Green function is recovered only if one takes a family whose retained sectors exhaust the full domain and whose width tends to zero:

$$K_{\text{coh}}(x, y) \rightsquigarrow \delta(x - y).$$

Remark 9.2 (Sector identity versus full identity). The distinction is crucial. A coherent projector may erase many upstream distinctions and still act as the identity on the retained effective variables. Thus K_{coh} is not “almost the identity” on the full upstream space. It is the exact or approximate identity on the downstream admissible sector.

Remark 9.3 (Physical compatibility gates). Even when K_{coh} is fixed mathematically, a physical substitution must preserve the structures used by the target theory. For a scalar elliptic problem these include covariance and boundary conditions. For gauge fields they also include the constraint complex and Ward or BRST identities. For Lorentzian propagation they include causal support and wavefront-set conditions. None of these follows from smoothness of the elliptic kernel alone.

10 Point particles, point charges, and finite source kernels

A point particle or point charge is usually encoded as

$$\rho(x) = q\delta(x - x_0).$$

The coherent replacement is

$$\rho_{\text{coh}}(x) = qK_{\text{coh}}(x, x_0),$$

or, in the heat-kernel model,

$$\rho_{\tau}(x) = qH_{\tau}(x, x_0).$$

The field equation becomes

$$L\varphi_{\text{coh}} = \rho_{\text{coh}},$$

with solution

$$\varphi_{\text{coh}}(x) = qG_{\text{coh}}(x, x_0).$$

In the heat model:

$$\varphi_{\tau}(x) = qG_{\tau}(x, x_0).$$

As $\tau \downarrow 0$, this converges to the standard point-source potential.

11 Contact interactions as coherent overlaps

A local contact interaction is often modeled by a zero-range kernel:

$$V(x, y) = g\delta(x - y).$$

The coherent replacement is

$$V_{\text{coh}}(x, y) = gK_{\text{coh}}(x, y).$$

For a quartic field interaction, the point-local expression

$$\lambda\phi^4(x)$$

can be replaced by an overlap vertex

$$\lambda \int_{X^4} V_{\text{coh}}(x_1, x_2, x_3, x_4) \phi(x_1) \phi(x_2) \phi(x_3) \phi(x_4) dx_1 \cdots dx_4,$$

where a simple coherent model is

$$V_{\text{coh}}(x_1, x_2, x_3, x_4) = \int_X K_{\text{coh}}(z, x_1) K_{\text{coh}}(z, x_2) K_{\text{coh}}(z, x_3) K_{\text{coh}}(z, x_4) dz.$$

This is the vertex-level analogue of the Green-function replacement:

$$\delta \rightsquigarrow K_{\text{coh}}.$$

12 Retarded and causal Green functions

The preceding sections used elliptic Green functions because the spectral theory is clean. Hyperbolic propagation requires causal support. The corresponding replacement is not

$$G_{\text{ret}}\delta$$

as a point impulse, but a finite-source retarded response:

$$LG_{\text{ret,coh}} = K_{\text{coh}}$$

with support controlled by the causal propagation of the finite source.

A basic model is

$$G_{\text{ret,coh}} = G_{\text{ret}} \circ \Pi_{\text{coh}}.$$

If G_{ret} has future-lightcone support and Π_{coh} is spatial/internal on a fixed time slab, the causal response is the retarded propagation of an admissible source rather than an acausal point impulse.

Remark 12.1 (Scope). A fully rigorous hyperbolic version requires specifying the spacetime function spaces, wavefront-set control, and causal support assumptions. This paper only records the structural replacement. The elliptic and heat-kernel theorems above are the rigorous core.

13 Diagnostics and finite-width effects

The coherent Green-function replacement suggests several concrete diagnostics.

1. **Finite-source models.** A supplied extended preparation or source can be represented by a finite kernel; point-source physics is not universally replaced.
2. **Finite diagonal response.** Coincident-point quantities are finite before the delta limit is taken.
3. **Modified contact scattering.** A declared nonlocal EFT interaction can acquire form factors determined by K_{coh} ; this is not implied by the free Green equation.
4. **Resolution-dependent propagators.** Propagators become $L^{-1}\Pi_{\text{coh}}$, not L^{-1} on all modes.
5. **Renormalization question.** A finite filter may soften selected integrals, but renormalization, Ward identities, reflection positivity, and Lorentzian unitarity require independent analysis.

14 Relation to the delta/projection paper

The preceding paper established the general diagnostic:

$$\text{find the delta} \quad \Rightarrow \quad \text{identify the hidden projection.}$$

The present paper executes the first instance:

$$LG = \delta \quad \Rightarrow \quad LG_{\text{coh}} = K_{\text{coh}}.$$

Thus the Green function is not abandoned. It is reclassified:

ordinary Green function = singular limit of coherent finite-source response.
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15 Conclusion

The Dirac delta enters Green-function theory as the ideal point source. A declared finite kernel gives a legitimate alternative source model and a controlled sharp limit. The mathematics alone does not decide which source is physically realized.

The rigorous theorem is simple: finite spectral Green functions $G_\Lambda = L^{-1}\Pi_\Lambda$ and heat-filtered Green functions $G_\tau = L^{-1}e^{-\tau\Delta}$ are smooth finite-source responses that converge distributionally to the ordinary Green function only in the singular limit. Their diagonal values are finite before that limit is taken.

This provides a worked conditional interface for the delta/projection principle. It turns the mathematical relation

$$\delta = \text{singular shadow of projection}$$

into the operational replacement

$$LG = \delta \quad \rightsquigarrow \quad LG_{\text{coh}} = K_{\text{coh}}.$$

The next step is not automatic substitution in other theories. It is to derive a selected kernel and prove the required covariance, gauge, causal, and interacting consistency conditions in each target domain.

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